

01 / DESIGN REPORT

EBEAM_xgragonk

Silicon MZI design for group-index extraction

Designer: kaitai xiao **Technology:** SiEPIC EBeam, TE at 1550 nm
GitHub username: xdragonk **edX username:** kaitaixiao

The objective is to extract the group index of a 500 nm × 220 nm silicon strip waveguide from the wavelength spacing of Mach-Zehnder interferometer (MZI) fringes. Five path-length differences provide different fringe spacings, while a grating-coupler loopback provides a reference for de-embedding the coupling envelope.

Design prediction at 1550 nm: $n_{\text{eff}} = 2.443187$, $n_g = 4.221244$. The layouts use $\Delta L = 30.000, 45.000, 60.000, 90.000, 120.000 \mu\text{m}$. The report includes native MODE graphs and executed INTERCONNECT compact-model spectra. Fabricated-device measurements remain future work.

Evidence and scope

Evidence	Used in this report
Electromagnetic MODE data	Actual guided-mode profile and exported wavelength-dependent n_{eff} data.
Compact model	Second-order least-squares fit to the exported effective index. Its derivative predicts group index.
Circuit simulation	Executed INTERCONNECT compact-model spectra, with an analytical MZI transfer function for comparison.
Physical design	Hierarchical PDK layout, routed waveguides, test labels, loopback and local verification.

Waveguide mode

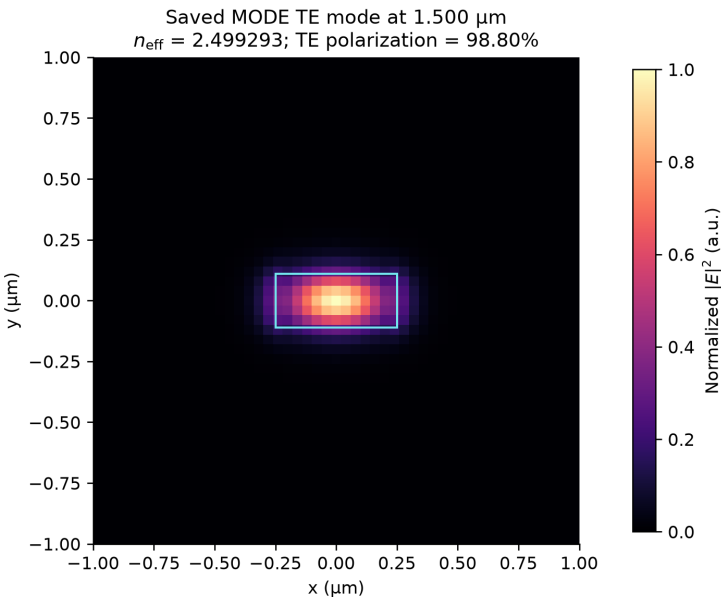


Figure 1. Actual saved MODE eigenmode at 1500 nm, replotted from its stored E field: normalized total intensity, $n_{\text{eff}} = 2.499292823$, TE fraction = 98.796%. Cyan marks the 500 × 220 nm Si core in SiO₂. The 2 × 2 μm, metal-boundary FDE uses a 51 × 51 field grid; mesh/domain convergence is not assessed in this draft.

02 / DESIGN REPORT

Dispersion and the compact waveguide model

The exported f column is in Hz. Divide by 10^{12} to obtain THz, then convert to vacuum wavelength: $\lambda(\mu\text{m}) = 299.792458 / f(\text{THz})$. The polynomial fits the real effective index against wavelength, with $x = \lambda - 1.55$ and λ in μm .

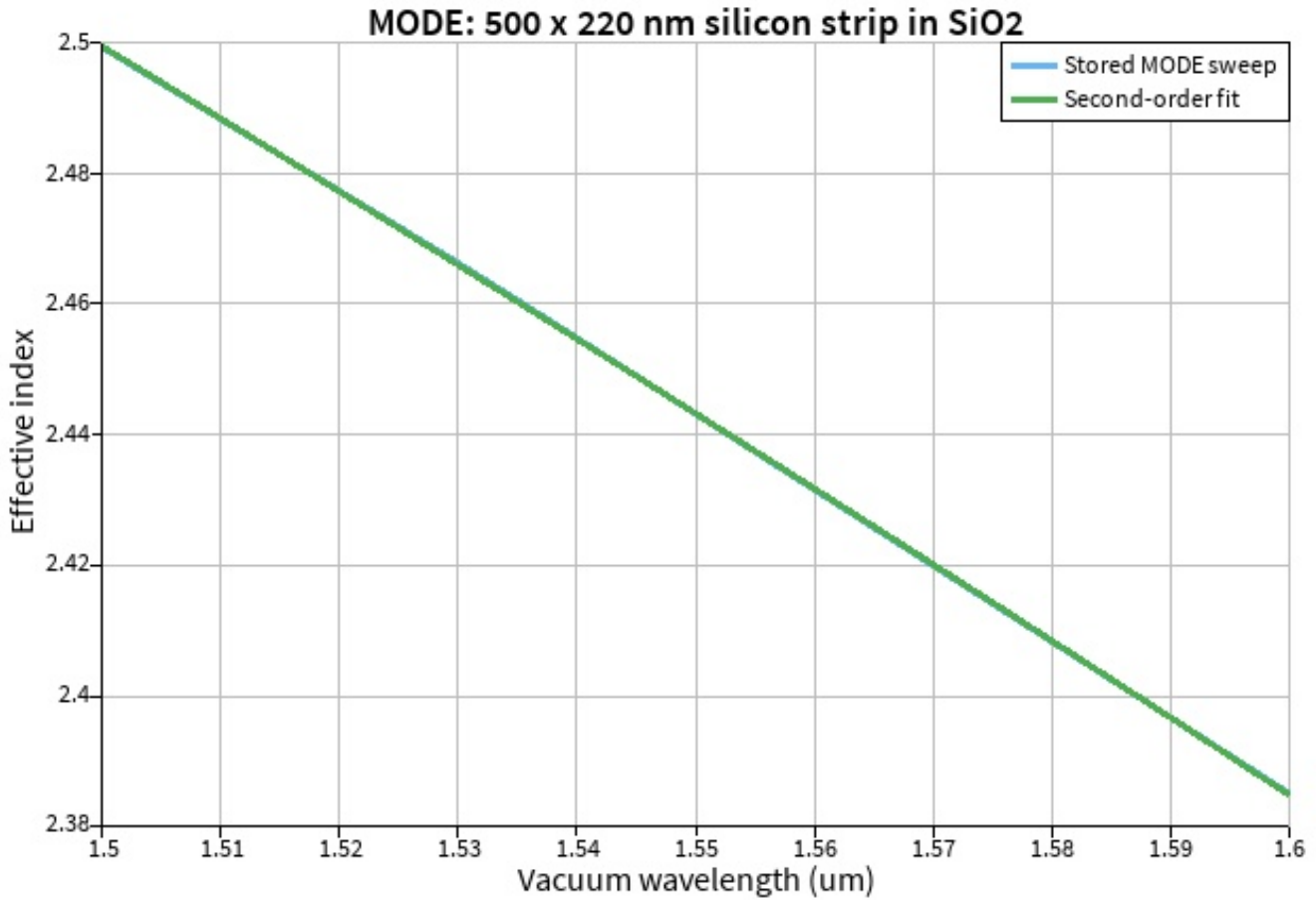


Figure 2. Native Lumerical MODE graph of the 20 stored n_{eff} samples and the quadratic fit. Sampling is uniform in frequency and therefore nonuniform in wavelength.

Second-order fit

$$n_{\text{eff}}(\lambda) = 2.44318723826 - 1.14713362515x - 0.439839400412x^2$$

$$dn_{\text{eff}}/d\lambda = -1.14713362515 - 0.879678800824x$$

$$n_g(\lambda) = n_{\text{eff}} - \lambda dn_{\text{eff}}/d\lambda$$

The model gives $n_g(1.55 \mu\text{m}) = \mathbf{4.221244357}$. Fit range: **1.50-1.60 μm** ; $R^2 = \mathbf{0.99998280}$; RMS n_{eff} residual = **1.44e-04**.

The coefficients have units 1, μm^{-1} , and μm^{-2} . This smooth derivative is used for the nominal MZI design. Its comparison with the original MODE derivative data is shown on the next page.

03 / DESIGN REPORT

Group-index and dispersion checks

Derivatives deserve a separate check because a small effective-index fitting residual can still produce a material difference in n_g and dispersion. The original MODE group-velocity export is converted through $n_g = c/v_g$.

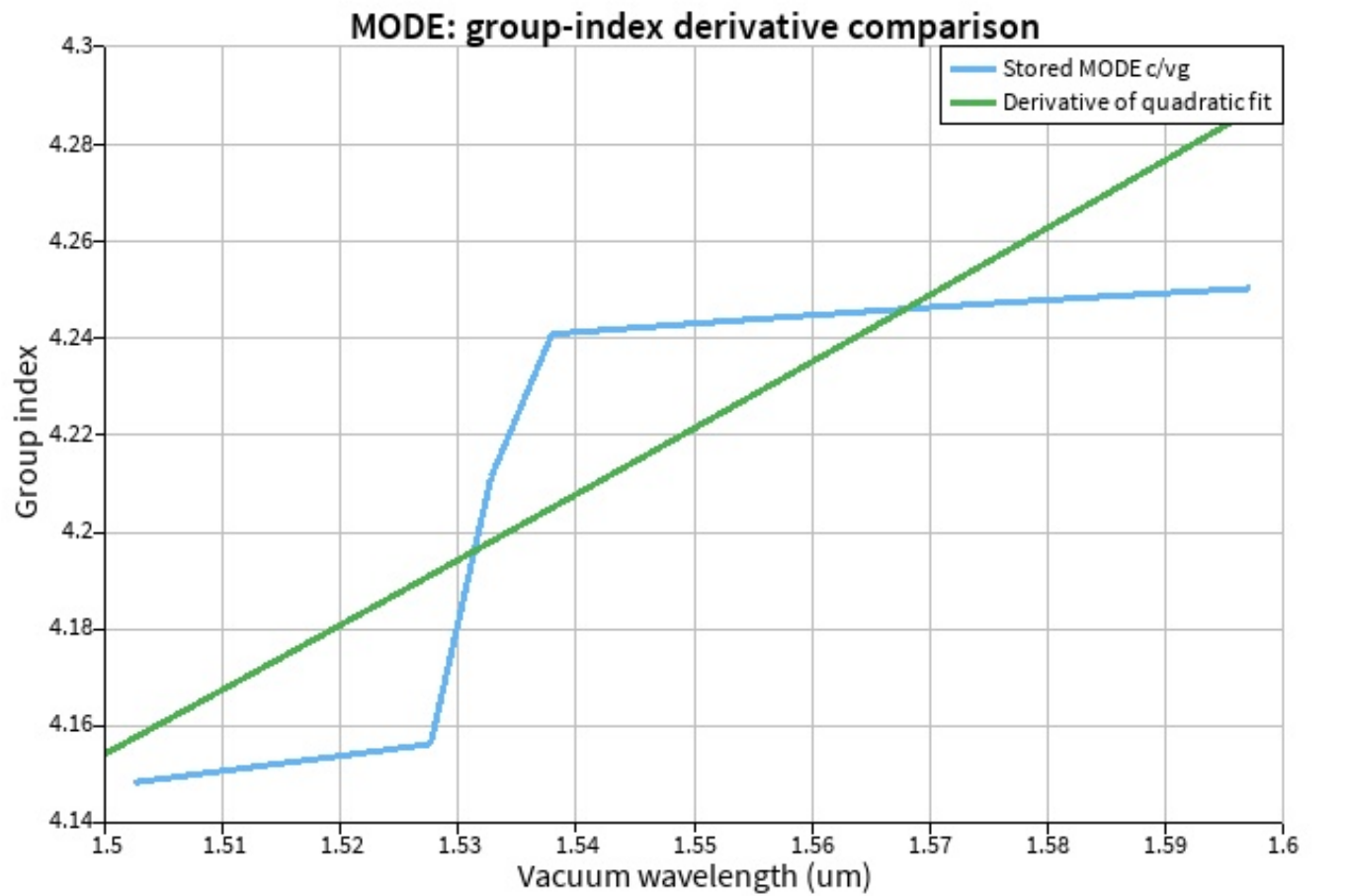


Figure 3. Native Lumerical MODE graph comparing the polynomial derivative with stored MODE c/v_g . The original data show a localized derivative change near 1.530-1.535 μm .

At 1550 nm	Value	Interpretation
Fitted n_g	4.221244357	Derivative of global second-order fit
MODE c/v_g	4.242847820	Linear interpolation of exported group index
Max n_g disagreement	0.859%	Over exported midpoint wavelengths

The global fit and MODE-derived n_g differ by up to 0.859%. Raw dispersion changes sharply near 1.530-1.535 μm ; its material or numerical origin is unverified. Exported v_g matches secant differences of exported beta, so that agreement is internal consistency. The high n_{eff} fit R^2 is not a mesh, material or dispersion-convergence result.

04 / DESIGN REPORT

Predicted MZI spectra and design variations

For two arms with physical path-length difference ΔL and the same waveguide cross-section, $\beta(\lambda) = 2\pi n_{\text{eff}}(\lambda)/\lambda$ and $\Delta\varphi = \beta\Delta L$. An ideal balanced splitter and combiner give the normalized output transmission below. Coupler insertion loss, propagation loss, splitter imbalance and reflections are omitted.

$$T(\lambda) = \frac{1}{2} [1 + \cos(\Delta\varphi)] = \cos^2[\pi n_{\text{eff}}(\lambda) \Delta L / \lambda]$$

$$\text{FSR}_\lambda \approx \lambda^2 / [n_g(\lambda) |\Delta L|]$$

Table 1. Nominal routed arm lengths and local FSR predictions.

Circuit	Short arm (μm)	Long arm (μm)	ΔL (μm)	FSR at 1550 nm (nm)
MZI01	165.307464	195.307464	30.000	18.9715
MZI02	165.307464	210.307464	45.000	12.6477
MZI03	165.307464	225.307464	60.000	9.4857
MZI04	165.307464	255.307464	90.000	6.3238
MZI05	165.307464	285.307464	120.000	4.7429

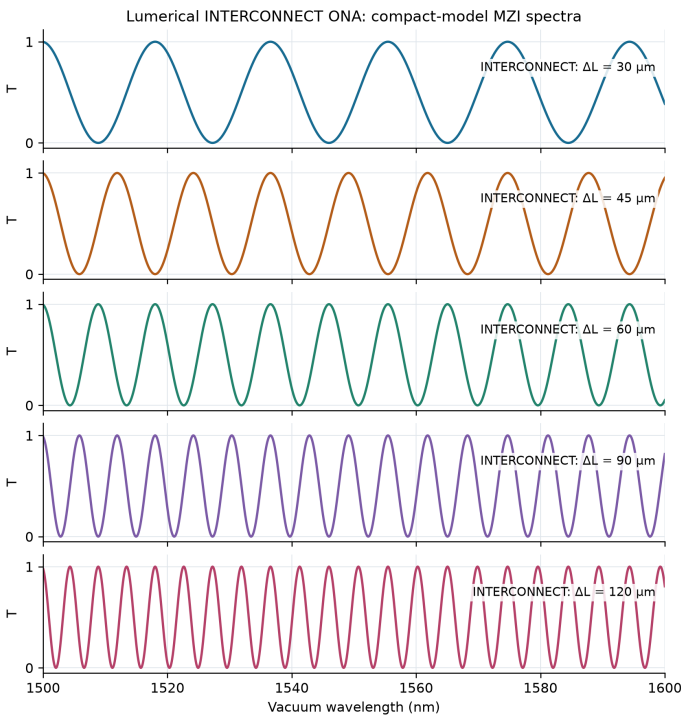


Figure 4. Five executed INTERCONNECT ONA spectra, replotted together to compare the path-length variations. All circuits use the final geometric ΔL . These are ideal compact-model circuit simulations; they omit measured component spectra and full-wave bend effects.

05 / DESIGN REPORT

INTERCONNECT circuit diagram

The native schematic below shows the compact-model circuit used for the MZI03 spectrum on the next page. ONA excites the circuit and measures its transmitted signal.

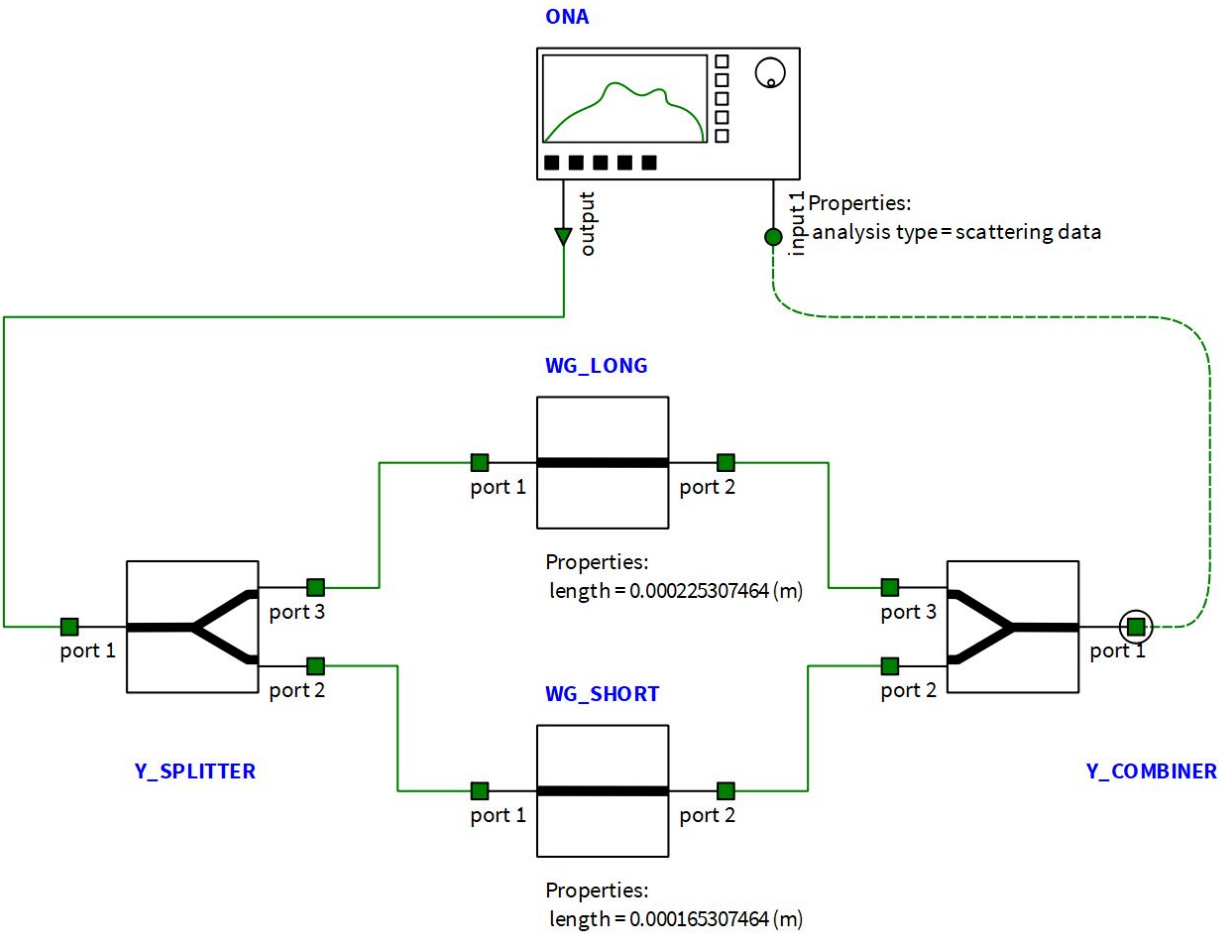


Figure 5. Actual Ansys Lumerical INTERCONNECT window capture of the executed MZI03 circuit, with display orientations arranged for readability and cropped to the schematic. WG_SHORT = 165.307464 μm ; WG_LONG = 225.307464 μm ; ΔL = 60 μm . The ideal 50:50 Y and waveguide ports are bidirectional. The dashed return to ONA.input 1 is a receiver connection, not resonant optical feedback.

Circuit connections and model scope

Path	Executed connection
Excitation	ONA output → Y_SPLITTER port 1.
Short arm	Splitter port 2 → WG_SHORT ports 1/2 → combiner port 2.
Long arm	Splitter port 3 → WG_LONG ports 1/2 → combiner port 3.
Measurement	Y_COMBINER port 1 → ONA input 1; scattering-data transmission.

The diagram represents two ideal 50:50 Y branches and equivalent straight-waveguide lengths. Grating-coupler spectra, bend spectra and measured component S-parameters are omitted. The other four runs keep this topology and vary only the long-arm length according to Table 1.

06 / DESIGN REPORT

Executed INTERCONNECT circuit simulation

Ansys Lumerical INTERCONNECT 2026 R1 was run for all five MZIs in a separate circuit project. Each circuit uses two bidirectional Waveguide Y Branch primitives and two Straight Waveguide equivalents with the routed arm lengths. The graph below is exported directly from INTERCONNECT.

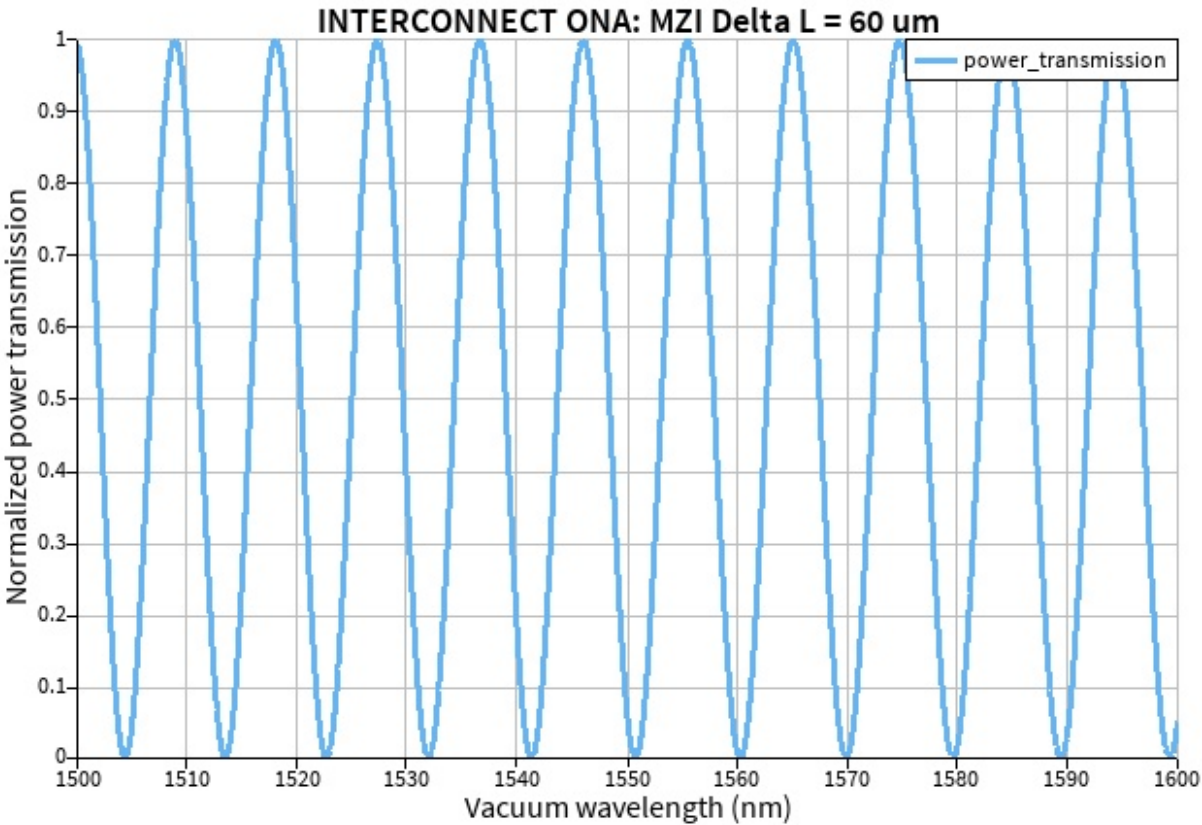


Figure 6. Native INTERCONNECT normalized output transmission for $\Delta L = 60\text{ }\mu\text{m}$, obtained from the executed optical network analyzer (ONA) scattering-data result.

Setting	Executed model
Splitter / combiner	Frequency-independent 50:50 Y branches; zero phase and modeled excess loss.
Waveguides	Short arm 165.307464 μm , long arm 225.307464 μm for this graph; zero propagation loss.
Reference at 1550 nm	$n_{\text{eff}} = 2.443187238$; $n_g = 4.221244357$; $D = 0.004548153581\text{ s/m}^2$.
Dispersion slope	$2934.292632624\text{ s/m}^3$; frequency-centered Taylor propagation-constant model.
ONA sweep	4001 uniform frequency samples spanning 1500-1600 nm; TE orthogonal identifier 1.

The largest normalized-power difference from the direct wavelength-polynomial calculation is 2.85×10^{-4} across these runs. This reflects compact-model truncation and parameter consistency. Both calculations use the same fitted dispersion; their agreement does not establish physical dispersion convergence.

07 / DESIGN REPORT

Layout implementation and test access

The physical layout uses unmodified EBeam PDK grating-coupler black-box cells and reciprocal Y-branches. Both arms of each MZI use the same strip cross-section and matching bend counts. This limits bend-related imbalance while the extra straight segment sets ΔL . Table 1 lengths are nominal PDK waveguide lengths, not fabricated measurements.

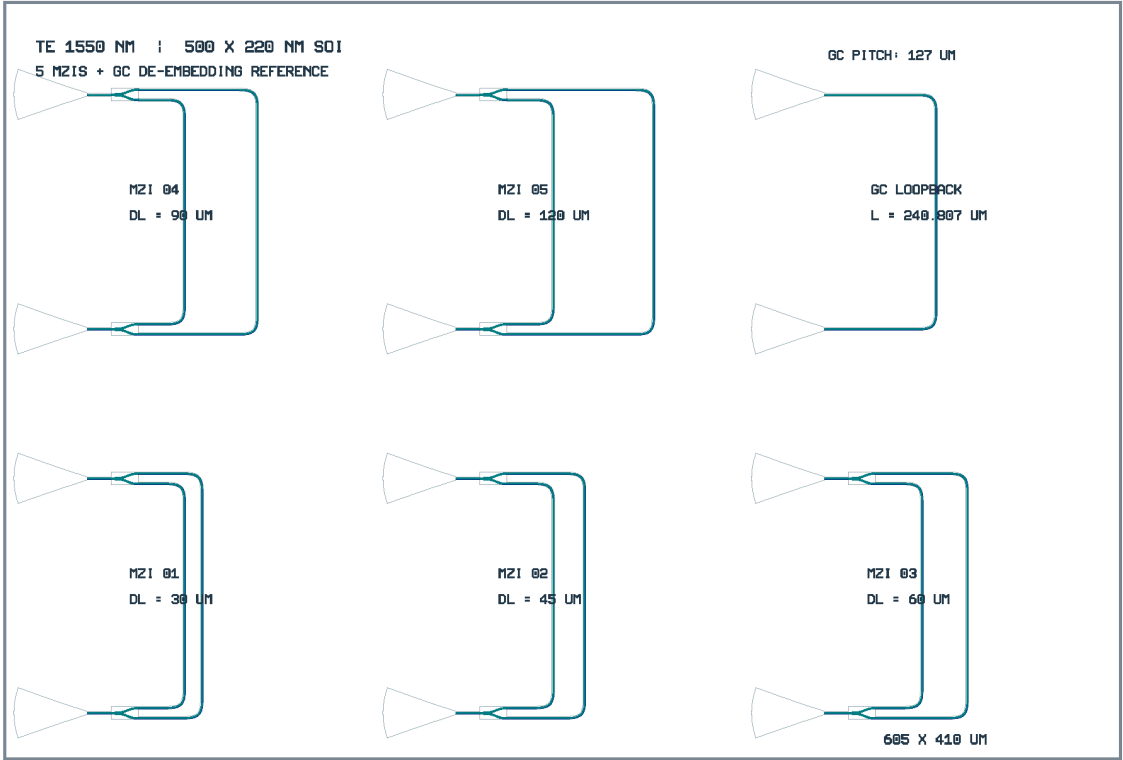


Figure 7. Native KLayout rendering of the final hierarchical layout with large annotations for readability. Top row: MZI04, MZI05 and loopback; bottom row: MZI01-MZI03. GC outlines identify black-box PDK cells; their foundry-defined internal geometry is retained for replacement at fabrication. Metadata text is hidden in this view only.

Item	Implementation
Die area	Floorplan 605 × 410 μm; physical geometry extent 477.550 × 341.000 μm.
Hierarchy	One top cell: EBeam_kaitaixiao. 5 MZI test circuits and one direct GC loopback.
Waveguides	500 nm strip; 220 nm device thickness; 10.0 μm radius parameter with PDK Bezier smoothing (0.20). Local curvature minimum was not separately evaluated.
Test access	GCs aligned vertically at 127 μm pitch; input opt_in label on Text 10/0 near the selected input GC.
Loopback	Reference connection between two matching GCs, used to remove the broad coupling envelope before fringe analysis.
Submission	EBeam_kaitaixiao.oas and EBEAM_xgragonk.pdf.

08 / DESIGN REPORT

Extracting group index from fringes

Why the FSR gives n_g

Successive transmission maxima differ in phase by 2π . Differentiating the MZI phase with respect to vacuum wavelength gives:

$$d\Delta\phi/d\lambda = 2\pi\Delta L [\lambda \, dn_{\text{eff}}/d\lambda - n_{\text{eff}}] / \lambda^2$$

$$d\Delta\phi/d\lambda = -2\pi\Delta L \, n_g / \lambda^2$$

For a small local fringe interval, $|d\Delta\phi/d\lambda|_{\text{FSR}} \approx 2\pi$. Thus $n_g \approx \lambda_c^2 / (|\Delta L|_{\text{FSR}_\lambda})$, using consistent length units and the midpoint λ_c between adjacent maxima.

Test on executed INTERCONNECT spectra

ΔL (μm)	Peak midpoint (nm)	FSR (nm)	Extracted n_g	Model n_g	Error
30.000	1546.020	18.897922	4.215943	4.215824	0.0028%
45.000	1555.488	12.714651	4.228793	4.228741	0.0012%
60.000	1550.722	9.492312	4.222259	4.222230	0.0007%
90.000	1552.300	6.337881	4.224395	4.224382	0.0003%
120.000	1548.344	4.735268	4.218996	4.218988	0.0002%

Consecutive maxima near 1550 nm are located in the actual 4001-point ONA results and refined by local fitting. Finite fringe spacing, peak localization and the frequency-Taylor model explain small differences from the fitted n_g . This checks the extraction calculation, not independent physical dispersion.

Experimental extraction procedure

1. Sweep a calibrated TE laser; record MZI and GC-loopback output in linear power.
2. Divide by a smooth loopback reference to suppress the GC envelope. Exclude low-signal reference regions and normalize.
3. Fit consecutive peak centers; calculate each local FSR and midpoint wavelength.
4. Calculate $n_g = \lambda_c^2 / (|\Delta L|_{\text{FSR}})$; compare all five MZIs and report wavelength, peak-fit and fabricated- ΔL uncertainty.

Fabrication changes the cross-section and n_g ; ΔL uncertainty biases extraction. Agreement across several ΔL values helps detect missed fringes and peak-fitting errors.

09 / DESIGN REPORT

Verification, limitations and references

Local verification results

Check	Result	Evidence
Submission structure	PASS, 0 errors	Official course validator: OAS and GDS each pass top-cell, black-box replacement, PDK layers and size checks.
Manufacturing DRC	PASS, 0 items	Local EBeam manufacturing DRC: boundary and physical geometry checks return zero markers.
Functional / connectivity	PASS, 0 errors	Reopened exports: 43 components, 42 optical nets, zero disconnected pins; course DFT check passes.
MZI geometry	PASS	All five ΔL values independently read from reopened extracted labels; equal bend counts and unmodified GC/Y cells.

Model and experiment limitations

The predicted spectra assume balanced, coherent components and exclude GC spectral response, radiation and propagation loss, reflections and process variation. The n_{eff} polynomial is an interpolation over 1.50-1.60 μm and should not be extrapolated. Derivatives are more sensitive than n_{eff} itself to mode-tracking changes and local sampling artifacts. The local checks establish geometric and functional consistency, not fabricated performance.

Submission status

The locally verified layout and this design proposal are prepared for the course draft submission. Native MODE index graphs and executed INTERCONNECT spectra are included. The source MODE and INTERCONNECT projects are retained for inspection. Fabrication and experimental characterization are future work.

References and source artifacts

[1] UBCx Phot1x 3T2026, Design Proposal - Report (course requirements). [edX assignment page](#)

[2] UBCx Phot1x 3T2026, Design Submission Checklist. [edX checklist](#)

[3] SiEPIC, openEBL October 2026 fabrication run and submission instructions. [Official fabrication repository](#)

[4] SiEPIC EBeam PDK, component library and waveguide definitions. [Official EBeam PDK](#)

[5] SiEPIC-Tools, functional layout verification and automated-test labels. [Official verification documentation](#)

[6] Ansys Lumerical, sweep analysis for dispersion calculations. [Ansys MODE documentation](#)

Numerical source: exported MODE table supplied for this assignment; frequency converted to wavelength before fitting. Layout and check results: final EBeam_kaitaixiao design artifacts generated with the installed SiEPIC EBeam PDK.